Spring 2022

Problem 1 (Fa88, Fa08, Sp22). Let f be a continuous, strictly increasing function from $[0, \infty)$ onto $[0, \infty)$ and let $g = f^{-1}$. Prove that

$$\int_0^a f(x)\,\mathrm{d}x + \int_0^b g(y)\,\mathrm{d}y \geqslant ab$$

for all positive numbers a and b, and determine the condition for equality. Use this to prove Young's Inequality, which states that if p and q are positive reals with $1/p + 1/q = 1$ and a and b are positive reals then

$$\frac{a^p}{p} + \frac{b^q}{q} \geqslant ab$$

Problem 2 (Sp22, Sp26). Suppose $f : [0, 1] \to \mathbb{R}$ is a continuous function $\displaystyle\int_0^1 x^n f(x)\,\mathrm{d}x = 0$ for all integers $n \geqslant 1$. Prove that f is identically 0.

Problem 3 (Sp22). Suppose that X is an uncountable subset of the reals. Prove that there is a point of X that is a limit of a sequence of distinct points of X.

Problem 4 (Sp22).

1. Show that there is a function $f(z)$, holomorphic near $z = 0$, such that

   $$f(z)^{10} = \frac{1}{\cos\left(z^5 + 2z^7\right)} - 1$$

   for all z in a neighborhood of $z = 0$.

2. Find the radius of convergence of its power series about $z = 0$. Your answer may involve a zero of an explicitly given polynomial.

Problem 5 (Sp22). Let $f(z)$ be a function holomorphic on the whole complex plane $\mathbb{C}$ such that $f(z) \in \mathbb{R}$ for all $z \in \mathbb{R}$. Show that $\overline{f(z)} = f(\bar{z})$ for all $z \in \mathbb{C}$.

Problem 6 (Sp81, Sp22). Let T be a linear transformation of a vector space V into itself. Suppose $x \in V$ is such that $T^m x = 0$, $T^{m-1}x \neq 0$ for some positive integer m. Show that $x, Tx, \ldots, T^{m-1}x$ are linearly independent.

Problem 7 (Sp22). Suppose n is a positive integer and let f be the function $f(x) = (1, x, x^2, \ldots, x^{n-1})$ from $\mathbb{R}$ to $\mathbb{R}^n$, show that a hyperplane containing the points $f(1), f(2), \ldots, f(n)$ does not pass through the origin.

Problem 8 (Sp22). Let A be an abelian group. Suppose that $a \in A$ and $b \in A$ have orders h and k, respectively, and that h and k are relatively prime. Let r and s be integers. Show that if $ra = sb$ then $ra = sb = 0$.

Problem 9 (Sp84, Sp22). Let $\mathbb{F}$ be a field and let X be a finite set. Let $R(X, \mathbb{F})$ be the ring of all functions from X to $\mathbb{F}$, endowed with the pointwise operations. What are the maximal ideals of $R(X, \mathbb{F})$?

Problem 10 (Sp81, Sp22). Which of the following series converge?

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)!(3n)!}{n!(4n)!}$ 2. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1+1/n}}$ 3. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^{1+1/\log^2 n}}$

Problem 11 (Sp22). Suppose that f is a smooth function from the reals to the reals satisfying the differential equation

$$f'(x) = \sin(f(x))e^{-x^2}$$

Prove that f is bounded.

Problem 12 (Sp22). Let the function f be given by $f(x) = 0$ if x is irrational and $f(x) = 1/n^2$ if $x = m/n$ where m, n are coprime integers and $n > 0$. Show that there is a point where f is continuous but not differentiable.

Problem 13 (Sp81, Sp10, Sp19, Sp22). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{x \sin x}{\left(1 + x^2\right)^2} \, \mathrm{d}x$.

Problem 14 (Sp90, Sp22). Let the function f be analytic and bounded in the complex half-plane $\Re z > 0$. Prove that for any positive real number c, the function f is uniformly continuous in the half-plane $\Re z > c$.

Problem 15 (Sp90, Fa93, Sp13, Sp18, Sp19, Sp22).

1. If A is a matrix over an algebraically closed field of characteristic 0 with $A^n = I$ for some $n > 0$ show that A is diagonalizable.
2. Show that over any field of characteristic $p > 0$ there is a matrix A, with $A^n = I$ for some $n > 0$, that is not diagonalizable.

Problem 16 (Sp22). Find the number of conjugacy classes of complex 5×5 matrices such that all eigenvalues are 1.

Problem 17 (Su80, Sp12, Sp21, Sp22, Fa22). Let G be a finite group and H be a subgroup.

1. Show that the number of subgroups of G conjugate to H divides the index of H.
2. Show that if $G = \bigcup_{g \in G} gHg^{-1}$ then $G = H$.

Problem 18 (Fa17, Sp22). Let $p(x)$ be a polynomial of degree d with real coefficients. Assume that $p(n) \in \mathbb{Z}$ for any $n \in \mathbb{Z}$. Prove that all coefficients of $d!p(x)$ are integers.